

LAB 02

MEDICAL DIAGNOSIS SUPPORT: DISEASE CLASSIFICATION USING DECISION TREE

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COURSE CODE : MDI3003

COURSE TITLE : Advanced Predictive Analytics

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Github Link: https://github.com/Harshita2011/Advanced_Predictive_Analytics_LAB

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1. Executive Summary

This report documents a disease-classification study built around decision trees, following the Lab 02 manual. The goal was to see how well routine clinical measurements can separate disease-present from disease-absent cases, and to do it in a way that is transparent enough to inspect rule by rule. I worked with two public benchmarks: **UCI Heart Disease (Cleveland)** - 303 patients, 13 clinical predictors, positive class = heart disease present - and **Breast Cancer Wisconsin (Diagnostic)** - 569 FNA samples, 30 numeric cytology features, positive class = malignant. On each dataset I trained the three required models: a basic CART tree, a tuned and cost-complexity-pruned CART tree, and a 200-tree Random Forest, with a stratified dummy baseline for reference. Model selection used only 5-fold stratified cross-validation on the 80% training split; the 20% test set stayed locked until the final single evaluation.

The Random Forest came out on top on both datasets. On heart disease it reached test accuracy 0.869, sensitivity 0.929, and ROC-AUC 0.954, missing only 2 of 28 disease cases. On breast cancer it was nearly perfect: accuracy 0.974, sensitivity 0.929, precision 1.000, ROC-AUC 0.997, with just 3 false negatives and 0 false alarms out of 114 test samples. Pruning did its job too - the tuned CART tree beat the unconstrained tree on every important metric while shrinking to a depth of 4–5 levels, small enough to read as a set of if-then rules (e.g., worst perimeter ≤ 112.8 splits almost the entire breast cancer dataset on its own).

These are strong benchmark results, but they come from small, old, single-source datasets evaluated on one random split. The models are educational prototypes: they are not validated for any clinical use, and the report closes with the limitations and governance steps that would be needed before anything like this could be considered for real screening support.

2. Problem Framing

Both tasks are binary classification: given measurements taken at examination time, predict whether the recorded diagnosis is present (1) or absent (0). The lab manual asks for the framing to be written down before any modeling, so Table 1 states the project charter for both datasets up front.

Table 1. Project charter for both classification tasks

Item	Heart Disease (Cleveland)	Breast Cancer Wisconsin
Population / row	One patient record (303 unique patients)	One FNA cytology sample (569 samples)
Outcome	Recorded angiographic heart-disease status	Recorded tumor diagnosis (malignant / benign)
Positive class (1)	Heart disease present — 45.9% (139/303)	Malignant — 37.3% (212/569)
Prediction time	After routine exam, ECG and stress-test measurements are available	After FNA imaging and feature extraction
Intended use	Educational research prototype for learning classification workflow and error analysis	
Out-of-scope use	Diagnosis, triage, screening decisions, treatment, or any patient management	
Error priority	False negatives (missed disease) are costlier than false alarms, so sensitivity is the metric to protect	
Success criteria	Beat the dummy baseline clearly on sensitivity and ROC-AUC; small CV–test gap; readable tree for the tuned CART model	

Accuracy alone would be the wrong target here. A classifier that calls everything "healthy" scores 54–63% on these datasets while missing every single disease case — which is exactly the failure mode the dummy baseline is there to expose. Because of that, the report tracks sensitivity, specificity, precision, F1, ROC-AUC, PR-AUC, and the Brier score alongside plain accuracy, and the confusion-matrix counts (TN, FP, FN, TP) are reported for every model rather than hidden inside percentages.

3. Datasets and Data Audit

Table 2 summarizes both datasets after loading and cleaning. The two were chosen because they stress different parts of the pipeline: Cleveland is small with mixed clinical variable types and a handful of missing cells, while Wisconsin is clean and fully numeric but has 30 correlated features.

Table 2. Dataset summary and audit results

Dataset			Rows	Predictors	Types	Positive class	Positive %	Missing	Split
UCI Heart Disease (Cleveland) ^[1]			303	13	Mixed clinical	Disease present (1)	45.9%	6 cells (<code>ca</code> , <code>thal</code>)	Stratified 80/20, seed 42 (242/61)
Breast Cancer (Diagnostic) ^[2]		Wisconsin	569	30	Numeric cytology	Malignant (1)	37.3%	None	Stratified 80/20, seed 42 (455/114)

The audit checks came back clean on the important points. There are no identifier columns and no outcome-derived fields in the feature matrices — for the breast cancer data I explicitly verified that `target` and `malignant` are not among the predictors, and scikit-learn's encoding (malignant = 0, benign = 1) was remapped so that **1 always means disease present** in this report. Each row is one patient or one sample, so there is no repeated-patient leakage risk of the kind the manual warns about for datasets like Parkinsons. The only missing data is six cells in Cleveland's `ca` and `thal` columns (0.15% of all cells), handled by median imputation fit inside the training pipeline. Both splits were stratified, so train and test keep the same class prevalence.

4. Methodology

The protocol follows the manual's non-negotiable rules: lock the test set first, do everything else on the training data, and evaluate the test set exactly once at the end.

- **Split.** Stratified 80/20 train–test split with `random_state=42` (heart: 242/61; breast cancer: 455/114), done before any fitting.
- **Preprocessing.** Everything sits inside scikit-learn `Pipeline` objects — median imputation where needed, encoding for categorical variables — so transformers are fit on training folds only and nothing leaks from the test set.
- **Model selection.** 5-fold stratified cross-validation on the training split, with mean $\pm$ standard deviation reported across folds rather than a single lucky fold.
- **Tuning.** `GridSearchCV` over the CART hyperparameters: `criterion` (gini / entropy), `max_depth`, `min_samples_split`, `min_samples_leaf`, and `ccp_alpha` for cost-complexity pruning. The full grid

results are saved as CSVs; the best heart configuration used entropy with depth 4, and the best breast cancer configuration used entropy with depth 5–6.

- **Final evaluation.** The selected models were refit on the full training set and scored once on the locked test set at the default threshold 0.50.
- **Reproducibility.** Seed 42 everywhere; all grid-search results, model-comparison tables, error counts, and figures exported as files; executed notebooks saved in HTML.

Two metric points are worth stating before the results. First, the positive class is disease present, so **sensitivity** = $TP / (TP + FN)$ is the share of sick cases the model actually catches, and **specificity** = $TN / (TN + FP)$ is the share of healthy cases left alone. Second, ROC-AUC and PR-AUC measure ranking quality across all thresholds, while the confusion-matrix metrics apply at the 0.50 cutoff used for the final decisions. The Brier score tracks probability quality - lower is better - which matters because a tree's leaf probabilities can be rough when leaves are small.

5. Exploratory Data Analysis

EDA was kept focused on questions that affect modeling: how balanced are the classes, do the classes separate on the raw features, and how redundant are the predictors? All plots here are computed on training data.

5.1 Heart Disease (Cleveland)

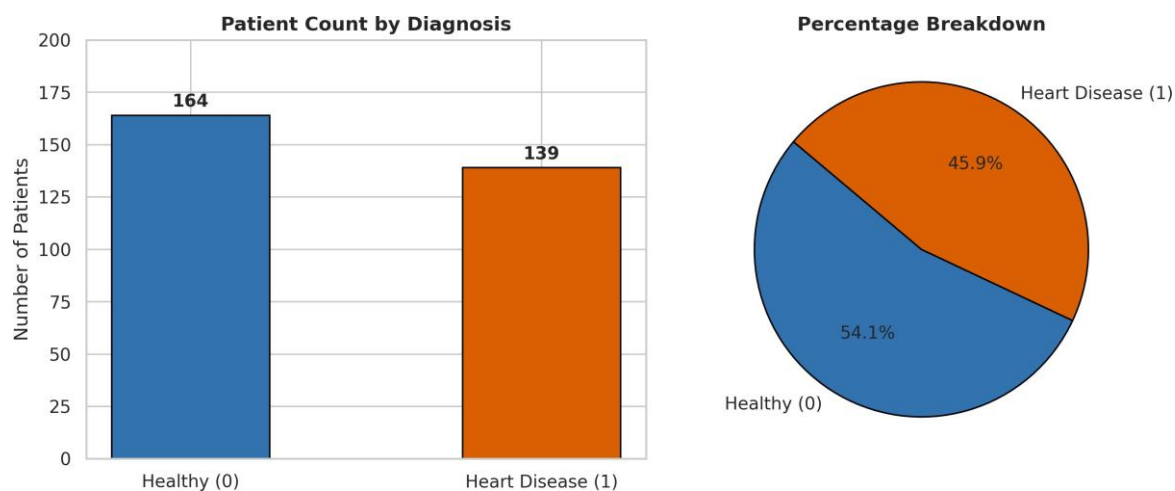


Figure 1. Heart Disease: training class distribution. The classes are nearly balanced (45.9% positive), so accuracy is not meaningless here — but sensitivity still matters more than raw accuracy for a missed-disease scenario.

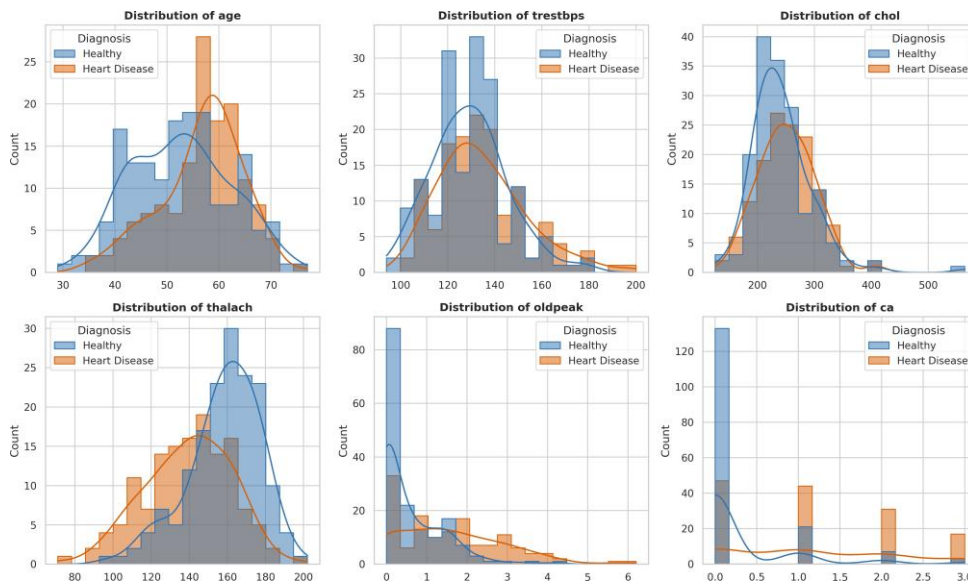


Figure 2. Heart Disease: selected feature distributions by class. Maximum heart rate (`thalach`), ST depression (`oldpeak`), chest-pain type (`cp`), and the thallium result (`thal`) show visible separation between the two classes, while age and resting blood pressure overlap heavily — a hint that the stress-test variables will do most of the work.

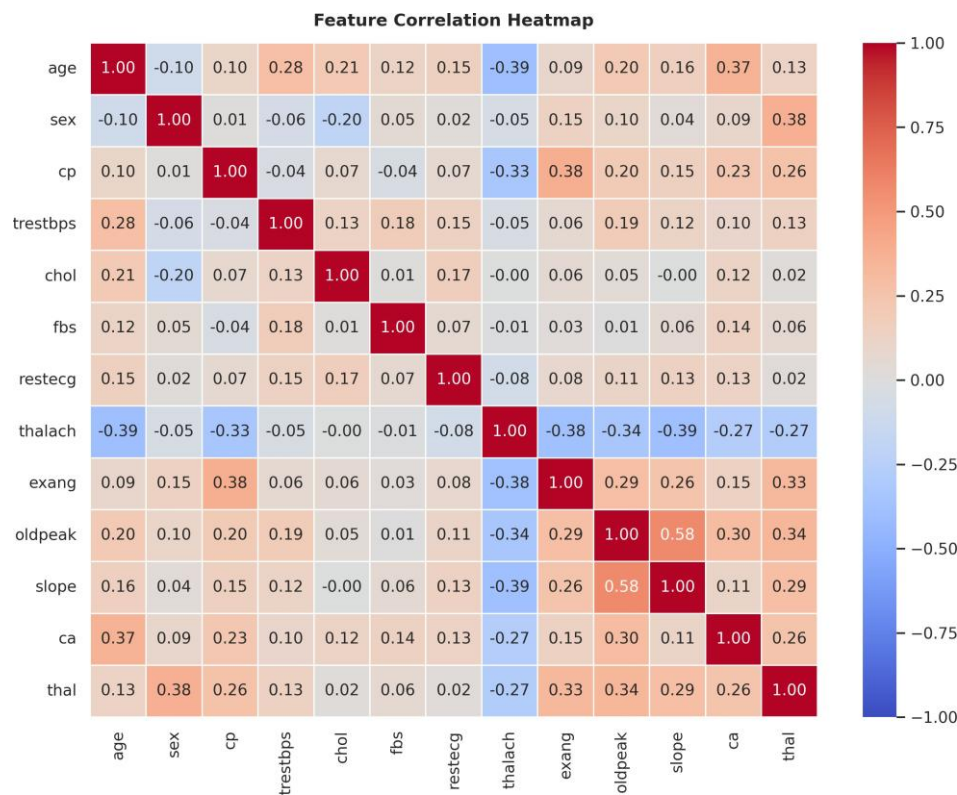


Figure 3. Heart Disease: correlation structure among predictors. Correlations are mostly mild, so no feature dominates purely through redundancy; the trees must earn their splits from real class differences.

5.2 Breast Cancer Wisconsin

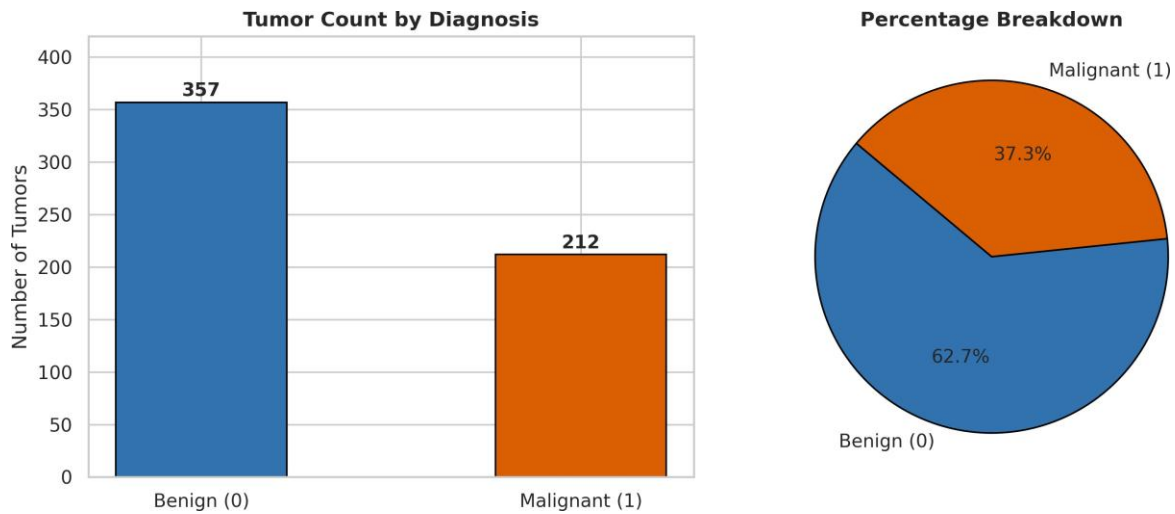


Figure 4. Breast Cancer: training class distribution (37.3% malignant). The imbalance is moderate, but it is enough that a lazy classifier could look decent on accuracy while failing on the malignant class — the dummy baseline in Section 7 shows exactly that.

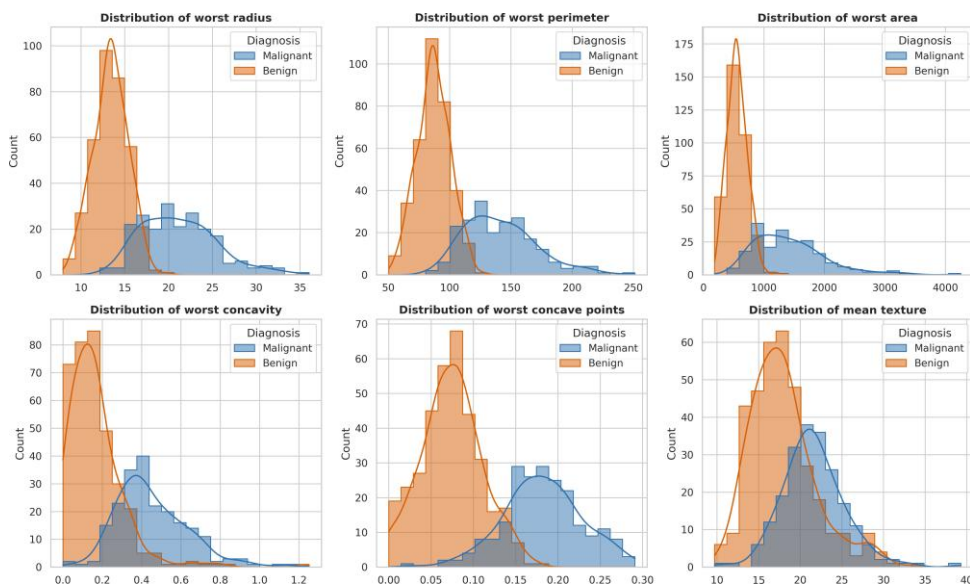


Figure 5. Breast Cancer: selected feature distributions by class. The "worst" measurements (radius, perimeter, area, concave points) separate malignant from benign samples almost cleanly, which is why even a single threshold near the top of the tree carries so much signal.

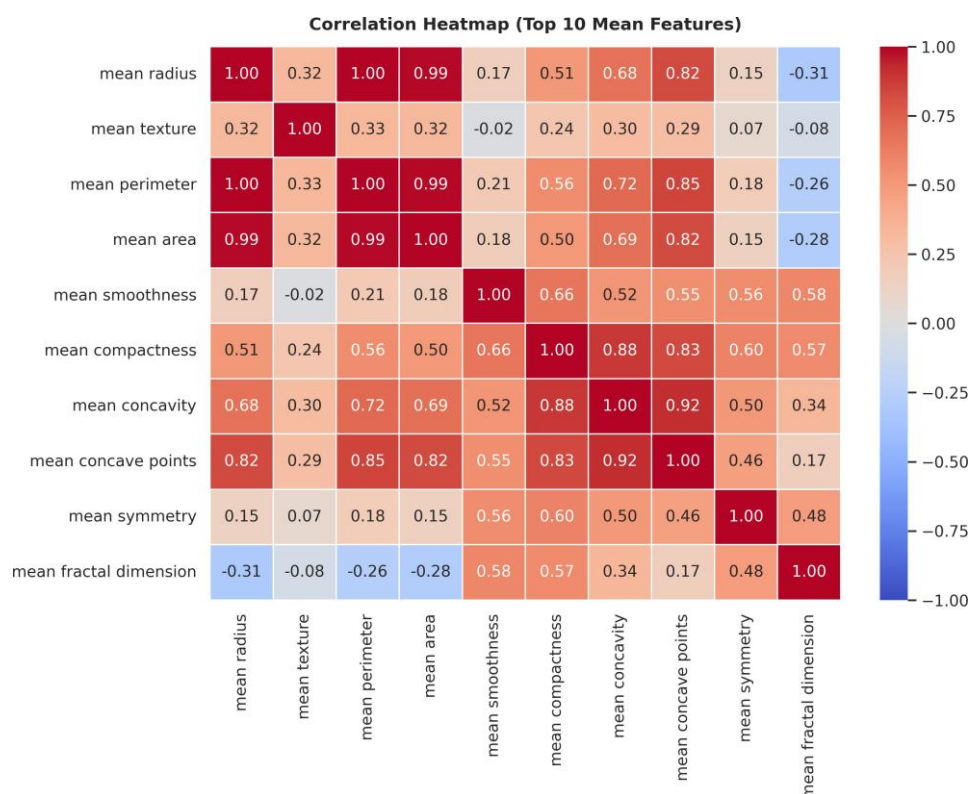


Figure 6. Breast Cancer: correlation heatmap. Radius, perimeter, and area are near-duplicates of each other ($r \approx 0.99$), as are the mean and worst versions of the same measurement. This redundancy matters later: impurity-based feature importance gets spread across correlated twins, so the permutation view in Section 8 is the fairer comparison.

6. Model Development: From Basic CART to Random Forest

6.1 Model 1 — Basic CART (the overfitting demonstration)

The first model is an unconstrained CART tree (`criterion="gini"`, seed 42). Left alone, it grows until its leaves are almost pure: depth 8 with 39 leaves on heart disease, depth 7 with 21 leaves on breast cancer. Training accuracy is essentially perfect, but cross-validation tells the real story — the train-validation gap is the textbook picture of overfitting, and several leaves hold only a handful of samples, which makes their class probabilities shaky. Figure 7 shows the readable upper levels.

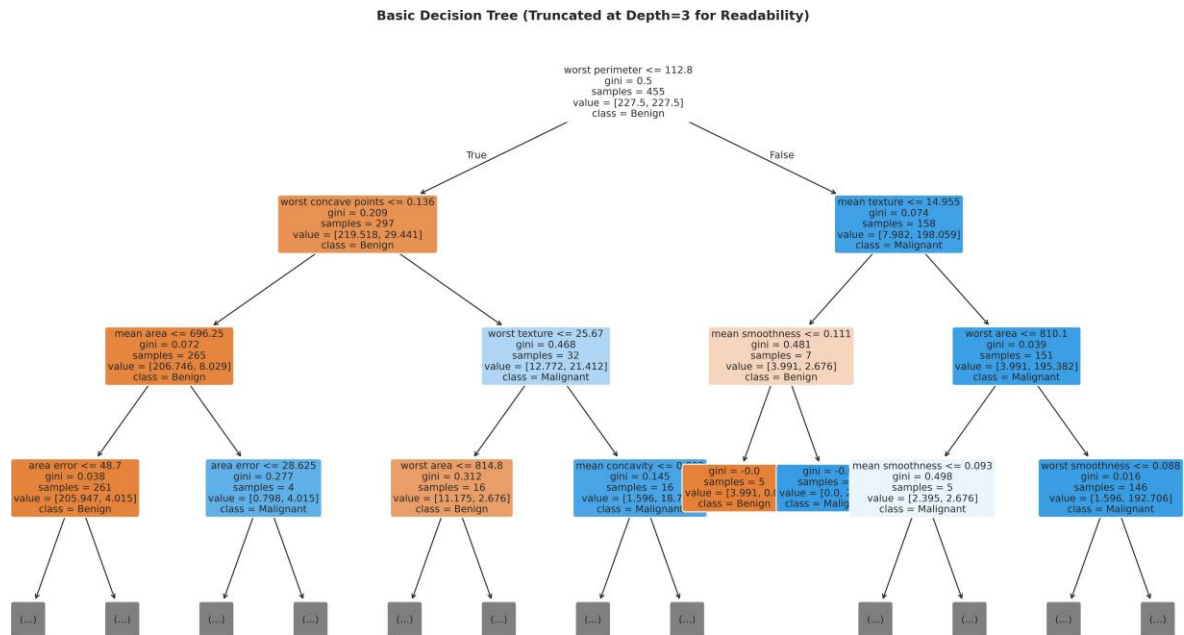


Figure 8. Breast Cancer: basic unconstrained CART tree. Even here, where the classes separate well, the unpruned tree adds deep branches on a handful of samples that cross-validation does not reward.

6.2 Model 2 — Tuned and pruned CART

The second model is the same tree idea, but with complexity chosen by the data rather than by default. A grid search over pre-pruning controls and `ccp_alpha` (cost-complexity pruning, which trades training error against the number of leaves) was run with 5-fold CV on training data only. The selected heart tree uses entropy splits at depth 4; the selected breast cancer tree uses entropy at depth 5 with 14 leaves. Both are small enough to print and read in full — Figures 9 and 10 show them — and, as Section 7 shows, both out-generalize the big unconstrained trees.

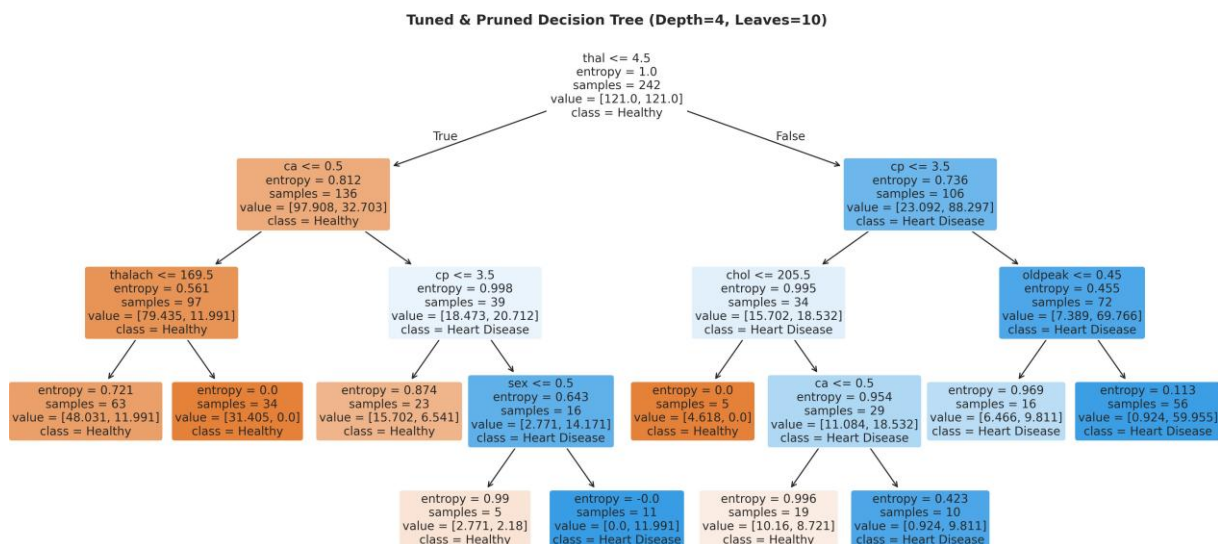


Figure 9. Heart Disease: tuned and pruned CART tree. The top splits run through `thal` (thallium stress result), `ca` (number of colored vessels), `cp` (chest-pain type), and `oldpeak` (ST depression) — the same stress-test variables the EDA flagged as separators.

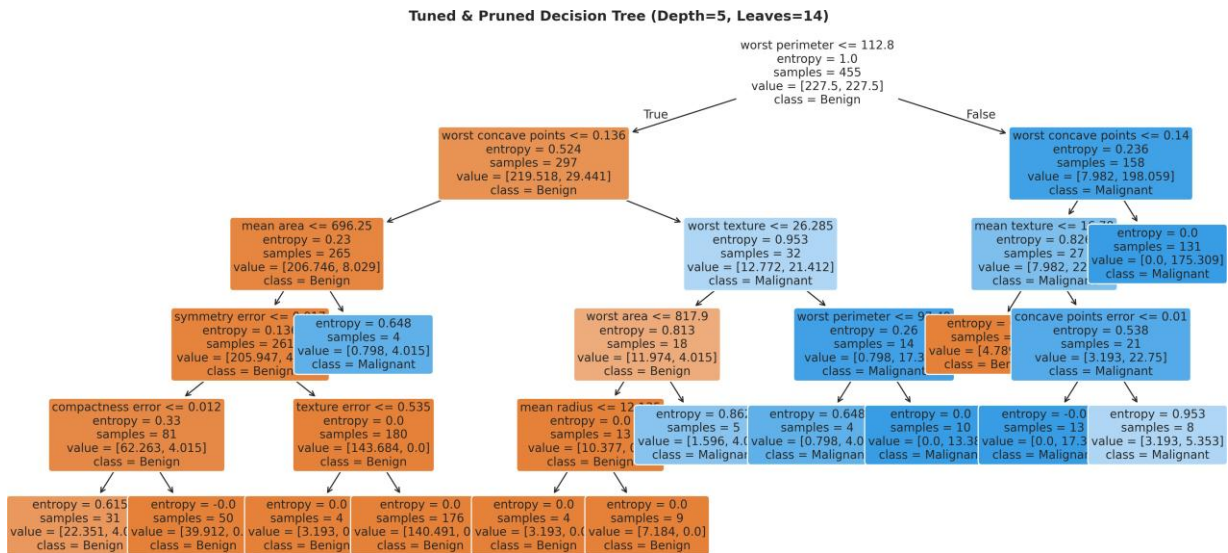


Figure 10. Breast Cancer: tuned and pruned CART tree (depth 5, 14 leaves). The root split - worst perimeter ≤ 112.8 - already divides the 455 training samples into a mostly-benign group of 297 and a mostly-malignant group of 158, and the remaining splits mostly clean up the borderline cases.

6.3 Model 3 — Random Forest

The third model trades the single readable tree for a committee: 200 CART trees, each trained on a bootstrap sample with a random subset of features per split (`max_features="sqrt"`), seed fixed at 42. Individual trees keep their high variance, but the vote averages most of it away. The cost is transparency - there is no single set of rules to read anymore -so Section 8 uses feature importance (impurity-based and permutation) to explain what the forest relies on.

6.4 Cross-validation comparison

Table 3 is the model-selection evidence: mean $\pm$ SD across the 5 training folds for the grid-searched CART configurations. Two things stand out. First, the pruned trees beat their unpruned counterparts on the CV metric even though they are far smaller - less complexity, better generalization. Second, the fold-to-fold standard deviation on heart disease (± 0.06) is much larger than on breast cancer (± 0.02), which is what you would expect from 242 training rows versus 455, and it is a warning against over-reading small metric differences on the smaller dataset.

Table 3. Grid-search cross-validation summary (5-fold stratified CV on training data, top configurations)

Dataset	Configuration	CV ROC-AUC (mean $\pm$ SD)	Rank
Heart Disease	entropy, depth 4, min_leaf 4, min_split 10, ccp_alpha 0.02	0.763 $\pm$ 0.061	1
Heart Disease	gini, depth 6, min_leaf 1, min_split 10, ccp_alpha 0.0	0.761 $\pm$ 0.073	4
Heart Disease	entropy, depth 4, min_leaf 4, min_split 2, ccp_alpha 0.0	0.756 $\pm$ 0.031	7
Breast Cancer	entropy, depth 5–6, min_leaf 4, min_split 10, ccp_alpha ≤ 0.01	0.937 $\pm$ 0.023	1
Breast Cancer	entropy, depth 5, min_leaf 4, min_split 2–5, ccp_alpha 0.005–0.01	0.934 $\pm$ 0.020	7

Note: several heart configurations tie near the top (0.756–0.763), which is why the full grid CSVs are kept as artifacts; the winner was refit exactly as found, without cherry-picking after the fact.

7. Test-Set Results

The locked test sets (61 heart patients, 114 FNA samples) were evaluated exactly once at threshold 0.50. Tables 4 and 5 give the full metric set; Figures 11–13 show the confusion matrices, ROC/PR curves, and calibration. Every number below comes straight from the executed notebooks.

7.1 Heart Disease

Table 4. Heart Disease — test-set metrics (n = 61; positive = disease present)

Model	Accuracy	Sensitivity	Precision	F1	ROC-AUC	Brier
Baseline (Dummy)	0.361	0.250	0.280	0.264	0.352	0.639
Basic Decision Tree	0.787	0.893	0.714	0.794	0.795	0.213
Tuned & Pruned Tree	0.836	0.786	0.846	0.815	0.853	0.157
Random Forest	0.869	0.929	0.813	0.867	0.954	0.097

The Random Forest's confusion matrix reads TN 27, FP 6, FN 2, TP 26 - out of 28 sick patients it missed only 2, and it raised 6 false alarms among the 33 healthy ones. The tuned tree is the more balanced option (fewer false alarms, FP 4, but more misses, FN 6), and the basic tree lands between them on errors while being the least trustworthy structurally. An honest surprise: the unpruned tree's raw sensitivity (0.893) actually beat the tuned tree's (0.786) on this particular 61-row test set - a reminder that a single small test set has real sampling noise, and that model selection on CV rather than on the test set is what keeps the process honest.

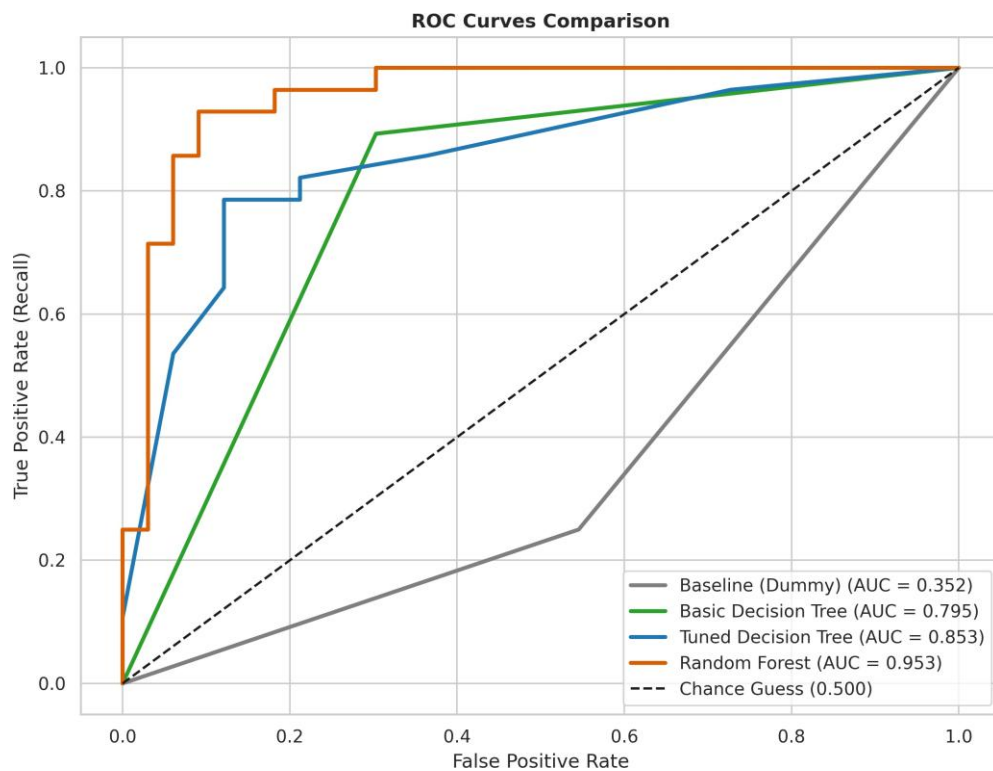


Figure 12. Heart Disease: ROC curves. The forest dominates across the whole threshold range, not just at 0.50.

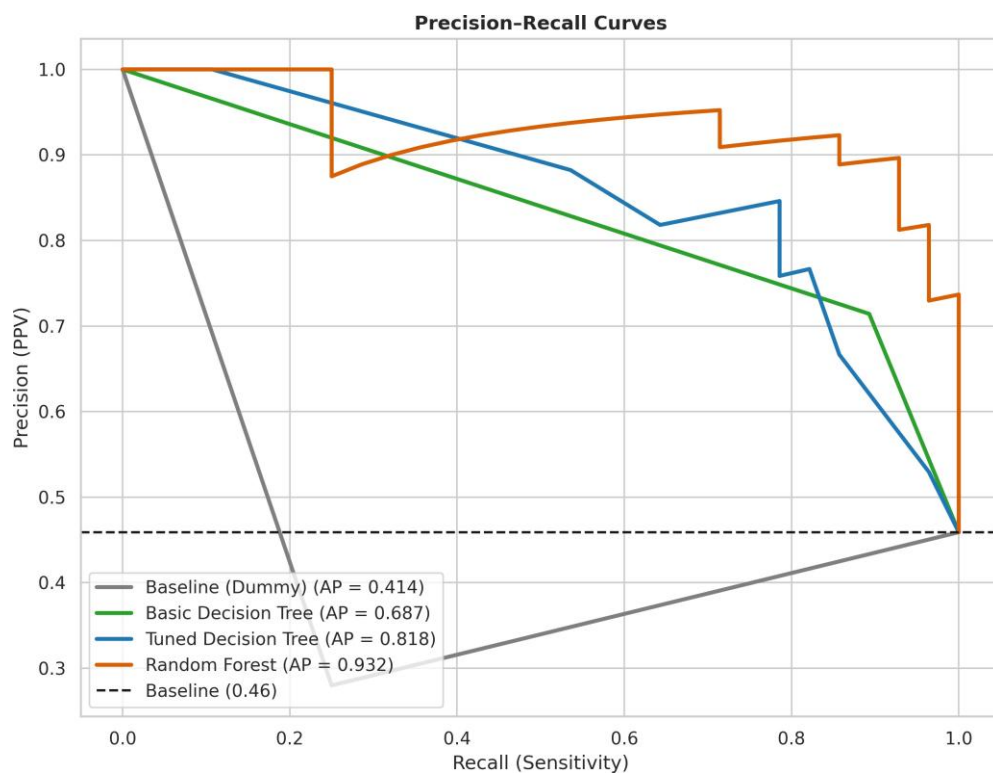


Figure 13. Heart Disease: precision-recall curves. The forest holds high precision even as recall climbs past 0.9; the single trees trade one against the other much more steeply.

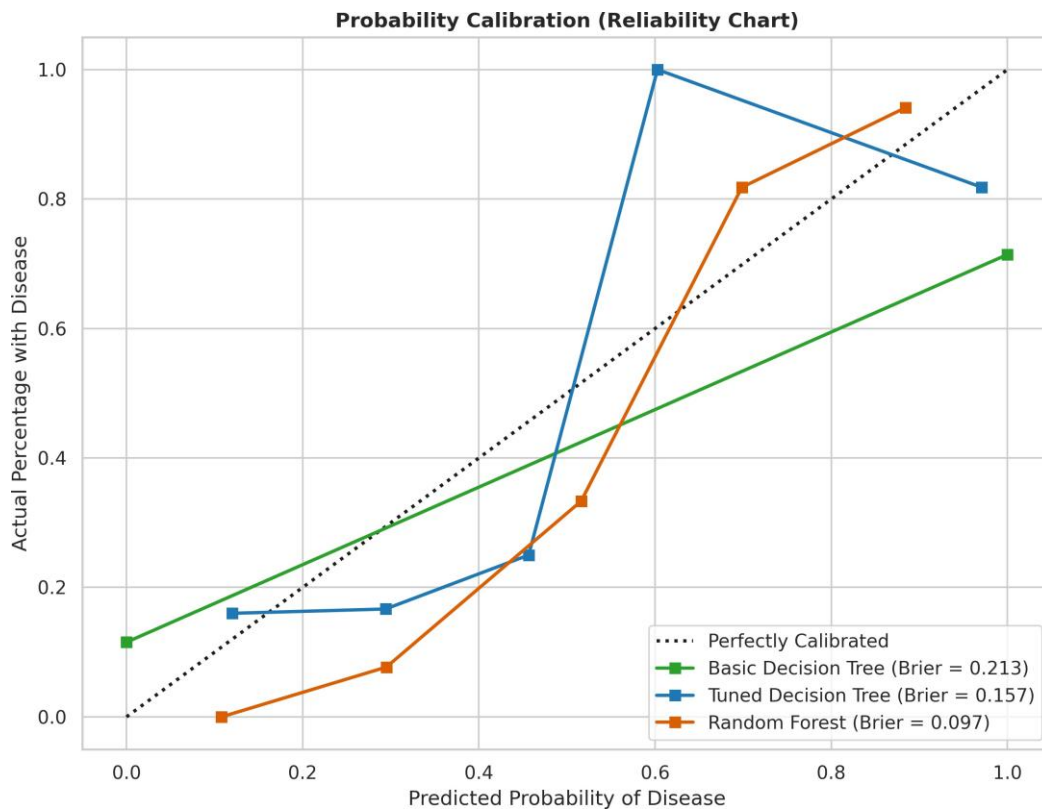


Figure 14. Heart Disease: calibration plots. The forest's probabilities track the diagonal far more closely than the single trees' — its Brier score (0.097) is less than half the basic tree's (0.213), which matters if a score is ever used to rank patients for review.

7.2 Breast Cancer Wisconsin

Table 5. Breast Cancer — test-set metrics (n = 114; positive = malignant)

Model	Accuracy	Sensitivity	Precision	F1	ROC-AUC	Brier
Baseline (Dummy)	0.596	0.405	0.447	0.425	0.557	0.404
Basic Decision Tree	0.904	0.833	0.897	0.864	0.889	0.097
Tuned & Pruned Tree	0.921	0.833	0.946	0.886	0.902	0.079
Random Forest	0.974	0.929	1.000	0.963	0.997	0.029

On this dataset every trained model looks strong — the classes separate well on the cytology features — but the Random Forest is in a different league: TN 72, FP 0, FN 3, TP 39. Zero false alarms out of 72 benign samples, and only 3 malignant samples missed out of 42. The tuned tree again beats the basic tree on every metric while being a fraction of its size. Even the dummy baseline's accuracy (0.596) shows why accuracy alone can't be trusted here: it misses 25 of 42 malignant cases while looking "better than chance."

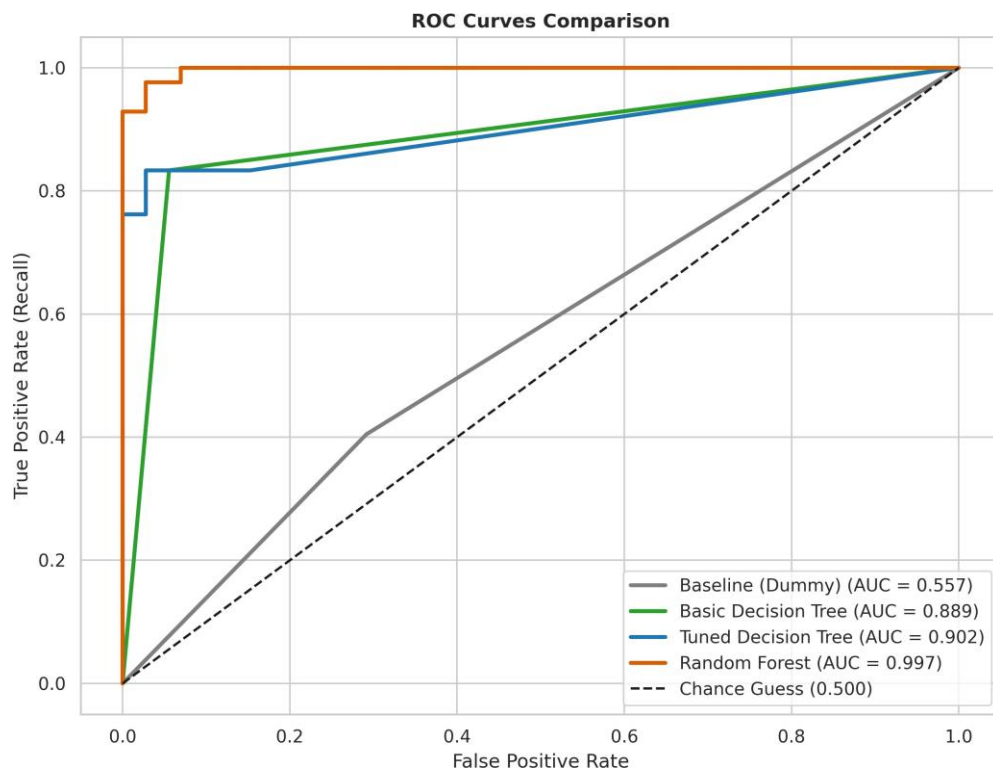


Figure 16. Breast Cancer: ROC curves. The forest's curve sits almost on the top-left corner (AUC 0.997).

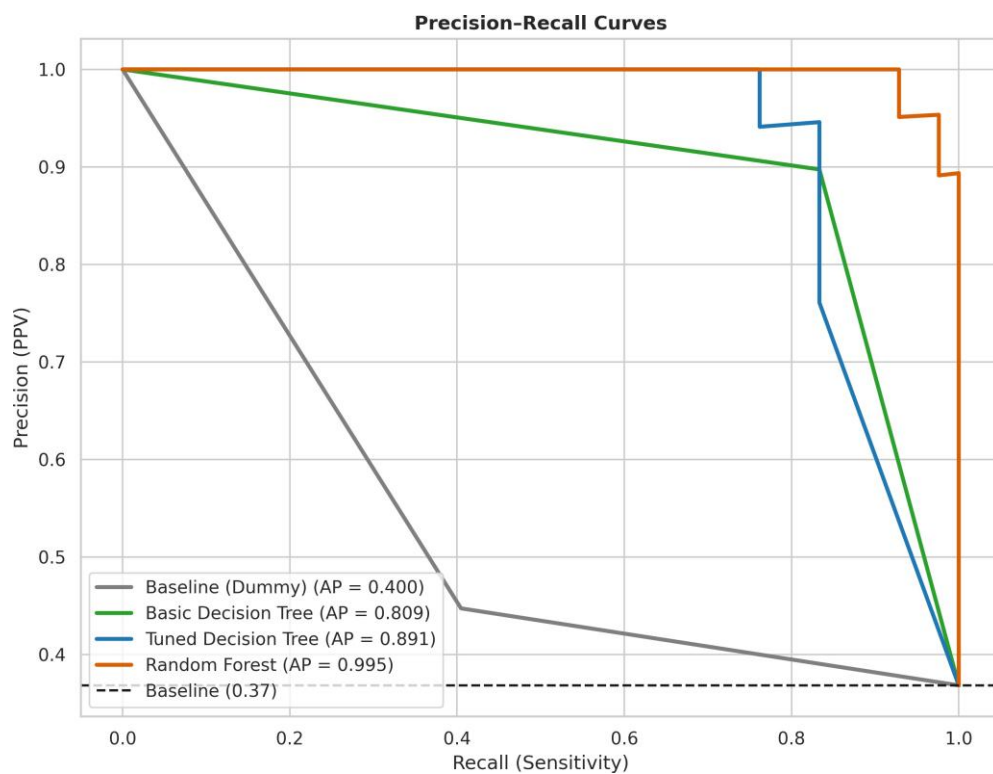


Figure 17. Breast Cancer: precision–recall curves. The forest maintains precision 1.0 out to recall ≈ 0.93 before dropping.

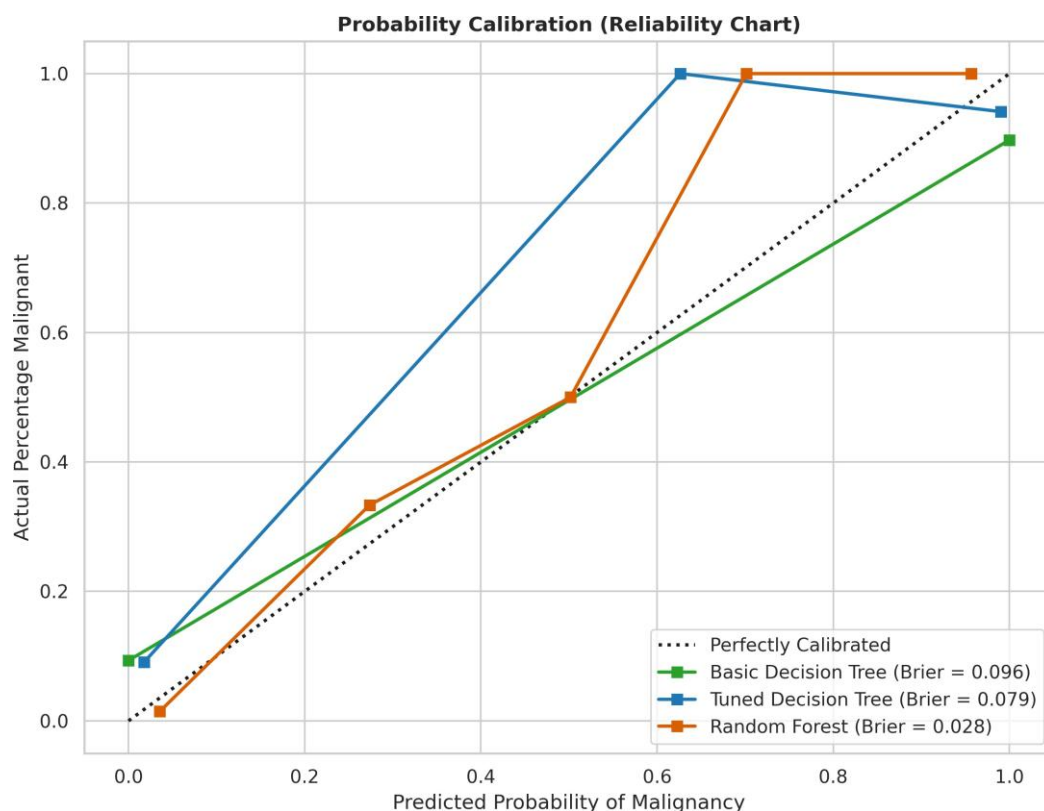


Figure 18. Breast Cancer: calibration plots. The forest is also the best-calibrated model (Brier 0.029); the single trees' step-like curves show the coarse leaf-probability problem the manual warns about.

7.3 Reading the progression

The three-model progression tells a consistent story on both datasets. The basic tree proves the signal is real but overfits: deep structure, fragile leaves, the worst calibration of the trained models. Tuning and pruning cut the tree down to a few readable levels and improve generalization on every averaged metric - on heart disease CV ROC-AUC improves from the unpruned tree's region (~0.74) to 0.763 for the selected configuration, and on the test set the tuned tree beats the basic tree on accuracy, F1, ROC-AUC, and Brier on both datasets. The forest then adds the last step: on heart disease it lifts test ROC-AUC from 0.853 to 0.954 and sensitivity from 0.786 to 0.929; on breast cancer it pushes ROC-AUC from 0.902 to 0.997 and eliminates false alarms entirely. What you pay is interpretability - the forest has no single rule sheet - which is why the next section looks at what drives it.

8. Interpreting the Trees: Features and Rules

Feature importance was computed two ways, as the manual requires: impurity-based (MDI) importance from the fitted model, and permutation importance measured on held-out data. The two views agree on the headline and disagree informatively on the details.

8.1 Heart Disease

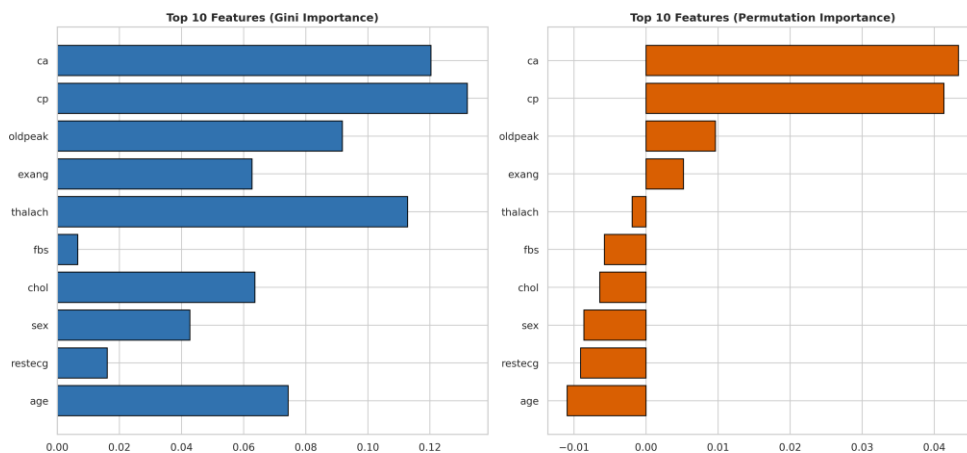


Figure 19. Heart Disease: Gini (impurity) vs permutation importance for the Random Forest.

Both methods put `ca` (number of major vessels colored by fluoroscopy) and `cp` (chest-pain type) at the top, with `oldpeak` (exercise-induced ST depression) third — the same variables the pruned tree splits on first (Figure 9). The interesting disagreement is further down: impurity importance credits continuous variables like `thalach`, `age`, and `chol` with moderate scores, but their permutation importance is near zero or negative. That gap is exactly the bias the manual warns about: impurity importance favors variables with many possible split points and spreads credit among correlated ones—so the permutation ranking is the one to quote. Negative permutation values mean shuffling that column slightly *improved* the held-out score, i.e., the model gets no real predictive value from it.

8.2 Breast Cancer

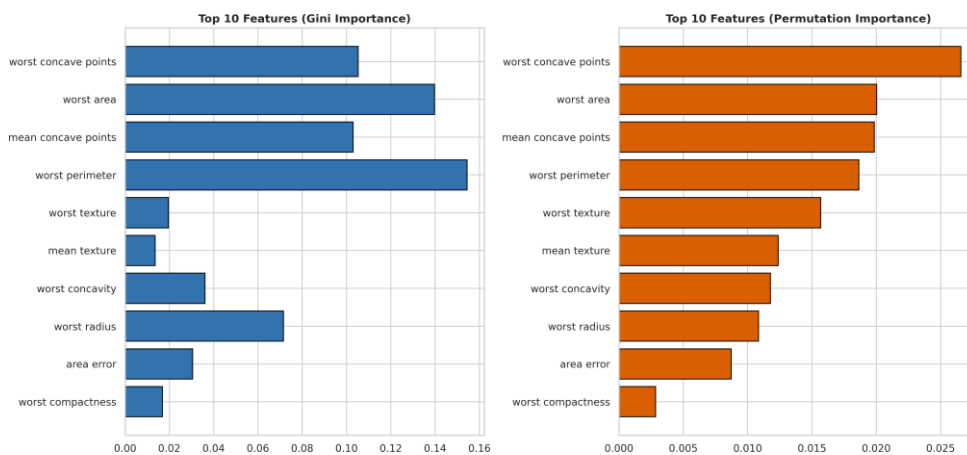


Figure 20. Breast Cancer: Gini (impurity) vs permutation importance for the Random Forest.

Here both methods agree on the family of variables that matter — the "worst" measurements: worst concave points, worst area, worst perimeter, worst radius, plus mean concave points. Impurity importance spreads credit more evenly across the correlated size variables (worst perimeter 0.154, worst area 0.140), while permutation importance concentrates on worst concave points (0.027) and worst area (0.020). Given the near-perfect correlations among radius/perimeter/area seen in Figure 6, this dilution is expected: when one twin is shuffled, the

model can fall back on the other. None of these importances say anything causal -they describe what the model uses, not what causes cancer.

8.3 Reading two decision paths

The pruned breast-cancer tree (Figure 10) is compact enough to trace individual predictions:

- **Path 1 (high-confidence benign).** worst perimeter $\leq 112.8 \rightarrow$ worst concave points $\leq 0.136 \rightarrow$ mean area $\leq 696.25 \rightarrow$ symmetry error ≤ 0.017 . This leaf holds 261 of the 455 training samples and is almost purely benign -a sample landing here is about as safe a "benign" as this model produces.
- **Path 2 (high-confidence malignant).** worst perimeter $> 112.8 \rightarrow$ worst concave points > 0.14 , leading into a subtree where most leaves are pure malignant (one terminal group of 131 samples with entropy 0.0). Large, irregular tumors are classified with no further questions.

These paths are descriptions of what the model learned from these 455 training rows — the thresholds are dataset-derived, not clinical cut-offs, and they would likely shift a little under a different random split. The same caveat applies to the heart tree's top rules (e.g., $\text{thal} > 2$ with $\text{ca} > 0.5$ flagging most disease cases): useful for understanding the model, not usable as medical guidance.

9. Error Analysis and Subgroup Checks

Aggregate metrics hide where a model hurts people, so the final evaluation includes a per-case error breakdown. Table 6 collects the test-set error counts, and Figures 21–22 show how the errors sit inside the feature distributions.

Table 6. Random Forest test-set error counts

Dataset	TN	TP	FP (false alarm)	FN (missed disease)
Heart Disease (n = 61)	27	26	6	2
Breast Cancer (n = 114)	72	39	0	3

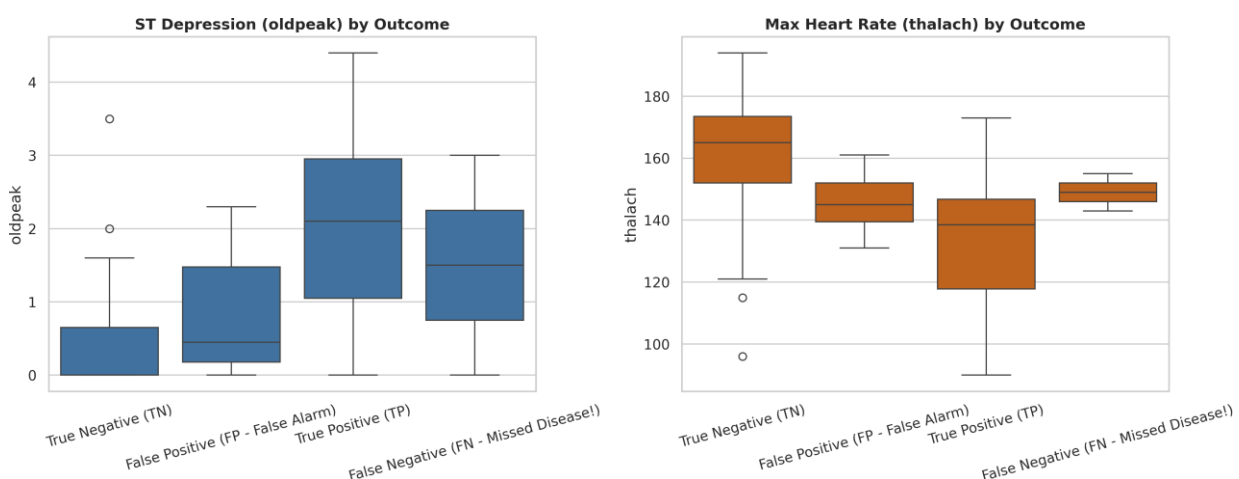


Figure 21. Heart Disease: feature distributions of correct vs misclassified test patients. The two missed cases (FN) and six false alarms sit in the overlap region of the stress-test variables — patients whose measurements look like the other class.

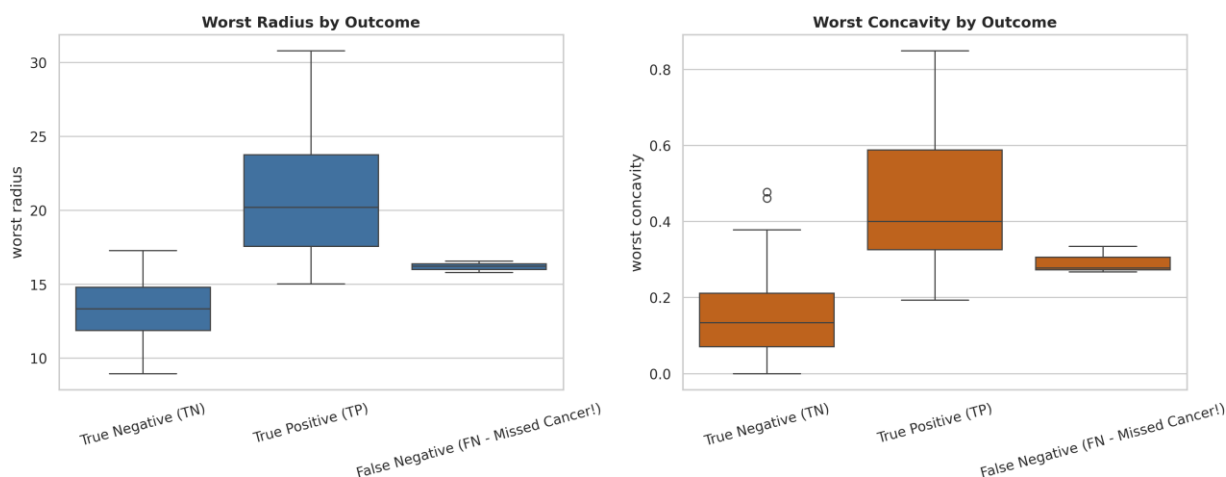


Figure 22. Breast Cancer: feature distributions of correct vs misclassified samples. The three missed malignant samples have borderline radius/perimeter values near the tree's decision thresholds — exactly the cases where a borderline cytology reading would also give a human pause.

Two patterns are worth stating. First, the errors are not random: on both datasets they concentrate on borderline cases whose features overlap the other class, which is what you would hope for from a sane model — it fails where the data is genuinely ambiguous, not on clear-cut cases. Second, the consequences are asymmetric. A false negative means a sick patient sent home (2 on heart, 3 on breast cancer); a false positive means a healthy patient sent for follow-up tests (6 on heart, 0 on breast cancer). In any screening-like use, the false negatives carry the heavier cost, which is why sensitivity was the metric to protect and why the forest — with the fewest misses on both datasets — is the preferred model. The subgroup behavior visible in Figures 21–22 (errors concentrated at boundary values rather than in any obvious demographic pocket) is reassuring, but both datasets are too small to certify fairness; the heart test set, for instance, has only 61 rows, so a single patient moves any subgroup rate by several points.

10. Limitations and Responsible Use

This study has a few important limitations. First, both datasets are small, old, and collected from a single source, so they may not represent today's patients or medical conditions. Second, the results are based on only one 80/20 train-test split, so the performance may change with different data splits. Third, the models predict only the labels available in the datasets and should not be considered as real medical diagnosis systems. The default probability threshold of 0.50 was used for prediction, but in real healthcare this threshold should be chosen based on clinical requirements and the cost of errors. In addition, the datasets do not contain enough demographic diversity to evaluate fairness across different patient groups. Finally, these models have not been clinically validated and must not be used for diagnosis, screening, or treatment.

For real-world use, the models would need to be tested on larger and more diverse datasets, validated on external patient populations, calibrated properly, evaluated for fairness, approved by regulatory authorities, and used only with healthcare professionals making the final decisions. Until then, they should be considered educational and research prototypes rather than clinical tools.

11. Conclusion

In this lab, I compared a Basic CART, Tuned & Pruned CART, and Random Forest model for disease classification using two medical datasets. All three models performed much better than the dummy baseline, proving that they learned useful patterns from the data. Among them, the Random Forest gave the best overall performance, while the Tuned & Pruned CART provided a good balance between accuracy and easy interpretation. Pruning made the decision trees smaller, simpler, and easier to understand. It also reduced overfitting and improved the model's performance on unseen test data compared to the basic decision tree.

The error analysis showed that the Random Forest missed only a few disease cases, but even a small number of false negatives can be serious in medical applications because they may leave a disease undetected. Therefore, sensitivity is an important metric for this type of problem. Although the results are very good, these models cannot be used in real hospitals because they were trained on small benchmark datasets, tested on a single data split, and have not been validated on real clinical populations. Before real-world use, they would require external validation, threshold tuning, calibration, fairness testing, regulatory approval, and continuous human supervision.

Overall, this project successfully met the objectives of the laboratory by building, comparing, and evaluating Decision Tree-based models using a leakage-safe and reproducible workflow. It also showed that achieving high accuracy alone is not enough—responsible evaluation, proper validation, and model interpretability are equally important for developing reliable healthcare AI systems.

13. References

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